UNIVERSIDAD DEL VALLE DE GUATEMALA

Redes — Sección 10

Laboratorio Práctico — Packet Tracer

Planificación IP, VLSM, VLANs y enrutamiento OSPF / EIGRP con redistribución

---

PDF 2

Evidencias de configuración y verificación

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Modalidad En parejas

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1. Topología final

Este documento reúne las salidas de los comandos de verificación tomadas en Packet Tracer con la configuración final: OSPF, EIGRP, redistribución mutua, balanceo de carga y sumarización. Como pide el enunciado, las salidas se presentan **en texto**, y la prueba de conectividad de extremo a extremo como **captura de pantalla**. Las líneas resaltadas son las más relevantes para cada criterio.

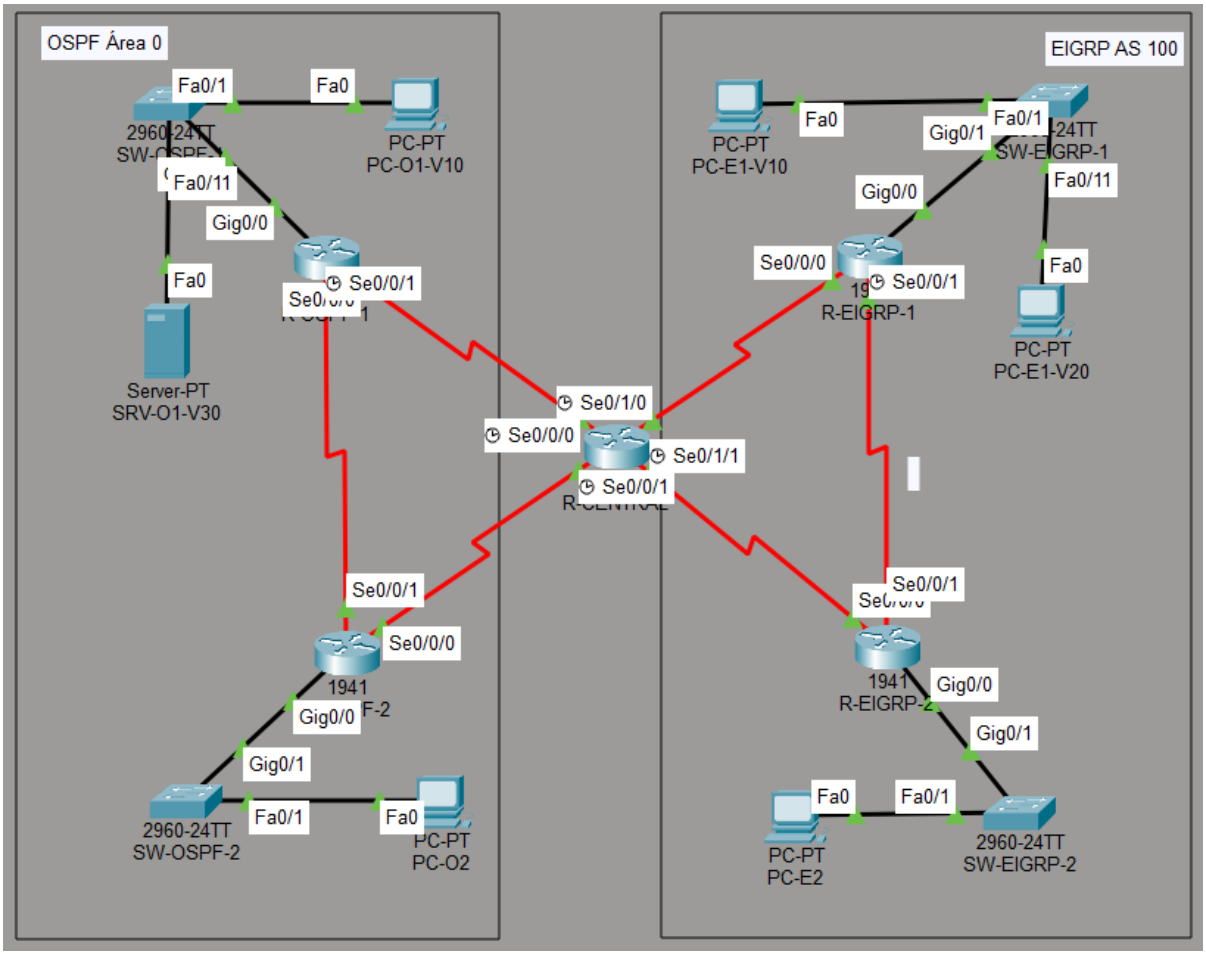


Figura 1. Topología final en Cisco Packet Tracer.

Criterio de la rúbrica	Dónde está la evidencia
Conectividad de extremo a extremo	Sección 5 (captura) y pruebas adicionales
Direccionamiento IP (VLSM)	PDF 1, secciones 3 y 4
Enrutamiento configurado (ruteo entre VLANs y redes anunciadas)	Sección 2 y anexos A y B
OSPF y balanceo de costo igual	Secciones 2.1, 2.2 y 3; anexo B
EIGRP y balanceo 75/25	Secciones 2.4 y 4; anexo B
Redistribución mutua	Sección 2 y anexo C
Sumarización de rutas	Sección 2 (rutas /23 y /28) y anexos C y D

2. Tablas de enrutamiento de los 5 routers

Salidas de `show ip route` tomadas después de configurar OSPF, EIGRP y la redistribución, con el balanceo de carga y la sumarización ya aplicados.

2.1 R-OSPF-1

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.1, 00:45:31, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
O E1    10.0.0.16/28 [110/84] via 10.0.0.1, 00:00:30, Serial0/0/0
     172.16.0.0/23 is subnetted, 1 subnets
O E1    172.16.0.0/23 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
O    192.168.0.0/24 [110/129] via 10.0.0.10, 00:45:31, Serial0/0/1
                    [110/129] via 10.0.0.1, 00:45:31, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

- **O E1 172.16.0.0/23** y **O E1 10.0.0.16/28:** todo el dominio EIGRP llega resumido en dos rutas externas de tipo 1, con métrica 20 (semilla) + 64 (serial hasta R-CENTRAL) = 84.
- **O 192.168.0.0/24** con dos caminos de costo 129 (por 10.0.0.10 y por 10.0.0.1): balanceo de costo igual (ECMP) hacia la LAN de R-OSPF-2.
- Las rutas **C** de las subinterfaces Gi0/0.10 y Gi0/0.30 son las VLAN 10 y 30 (router-on-a-stick).

2.2 R-OSPF-2

```
R-OSPF-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
O       10.0.0.0/30 [110/128] via 10.0.0.5, 00:01:02, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
O E1    10.0.0.16/28 [110/84] via 10.0.0.5, 00:01:02, Serial0/0/0
     172.16.0.0/23 is subnetted, 1 subnets
O E1    172.16.0.0/23 [110/84] via 10.0.0.5, 00:01:02, Serial0/0/0
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/129] via 10.0.0.9, 00:00:52, Serial0/0/1
                       [110/129] via 10.0.0.5, 00:00:52, Serial0/0/0
O       192.168.1.64/28 [110/129] via 10.0.0.9, 00:00:52, Serial0/0/1
                        [110/129] via 10.0.0.5, 00:00:52, Serial0/0/0
```

- **O 192.168.1.0/26** y **O 192.168.1.64/28** (VLANs de R-OSPF-1) con dos caminos de costo 129: por el enlace directo (10.0.0.9) y por R-CENTRAL (10.0.0.5). Es el ECMP en el sentido contrario.
- Los resúmenes externos **172.16.0.0/23** y **10.0.0.16/28** llegan por R-CENTRAL (10.0.0.5) con métrica E1 de 84.

2.3 R-CENTRAL

```
R-CENTRAL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
D       10.0.0.0/28 is a summary, 00:02:44, Null0
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/192] via 10.0.0.2, 00:47:26, Serial0/0/0
                    [110/192] via 10.0.0.6, 00:47:26, Serial0/0/1
S       10.0.0.16/28 is directly connected, Null0
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.17/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.21/32 is directly connected, Serial0/1/1
D       10.0.0.24/30 [90/2937856] via 10.0.0.18, 00:02:42, Serial0/1/0
                     [90/2937856] via 10.0.0.22, 00:02:42, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
S       172.16.0.0/23 is directly connected, Null0
D       172.16.0.0/25 [90/2170112] via 10.0.0.22, 00:02:42, Serial0/1/1
D       172.16.0.128/26 [90/2172416] via 10.0.0.18, 00:02:42, Serial0/1/0
D       172.16.0.192/26 [90/2172416] via 10.0.0.18, 00:02:42, Serial0/1/0
D       172.16.1.0/28 [90/2172416] via 10.0.0.18, 00:02:42, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:19:41, Null0
O    192.168.0.0/24 [110/65] via 10.0.0.6, 01:39:44, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.2, 01:40:09, Serial0/0/0
O       192.168.1.64/28 [110/65] via 10.0.0.2, 01:40:09, Serial0/0/0
```

- Es el único router que conoce **todas las rutas específicas** de ambos dominios: **O** para el lado OSPF y **D** para el lado EIGRP.
- **Cuatro rutas hacia Null0:** los dos resúmenes EIGRP (`D ... is a summary`: 192.168.0.0/23 y 10.0.0.0/28) y las dos estáticas que se redistribuyen en OSPF como resumen (`S`: 172.16.0.0/23 y 10.0.0.16/28). Las rutas específicas ganan por ser más largas, así que el tráfico real nunca termina en Null0.
- **O 10.0.0.8/30 [110/192]** con dos caminos (64 + 128 por cualquiera de los dos lados) y **D 10.0.0.24/30 [90/2937856]** = 256 × (6,476 + 2,000 + 3,000), también con dos caminos iguales.

2.4 R-EIGRP-1

```
R-EIGRP-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
D       10.0.0.0/28 [90/7264256] via 10.0.0.17, 00:08:27, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/0
L       10.0.0.18/32 is directly connected, Serial0/0/0
D       10.0.0.20/30 [90/7264256] via 10.0.0.17, 00:08:27, Serial0/0/0
                     [90/7520256] via 10.0.0.26, 00:37:50, Serial0/0/1
C       10.0.0.24/30 is directly connected, Serial0/0/1
L       10.0.0.25/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 7 subnets, 4 masks
D       172.16.0.0/25 [90/2426112] via 10.0.0.26, 00:37:50, Serial0/0/1
                      [90/7264512] via 10.0.0.17, 00:08:27, Serial0/0/0
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/26 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
C       172.16.1.0/28 is directly connected, GigabitEthernet0/0.99
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0.99
D    192.168.0.0/23 [90/7264256] via 10.0.0.17, 00:08:27, Serial0/0/0
```

- **D 192.168.0.0/23** y **D 10.0.0.0/28:** el lado OSPF llega resumido. Aparecen como rutas internas (AD 90) porque EIGRP anuncia los resúmenes como internos; la métrica 7,264,256 = 256 × (6,476 + 19,900 + 2,000) corresponde a la semilla más el camino hasta R-CENTRAL.
- **D 172.16.0.0/25** con dos caminos de **métrica distinta**: 2,426,112 (enlace directo) y 7,264,512 (por R-CENTRAL). Es el balanceo de costo desigual (variance), con una proporción de 2.994 a 1.
- **D 10.0.0.20/30** también usa dos caminos desiguales, como efecto de `variance 3` sobre toda ruta que cumpla la condición de factibilidad.

2.5 R-EIGRP-2

```
R-EIGRP-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
D       10.0.0.0/28 [90/7264256] via 10.0.0.21, 00:02:15, Serial0/0/0
D       10.0.0.16/30 [90/7264256] via 10.0.0.21, 00:02:15, Serial0/0/0
                     [90/7520256] via 10.0.0.25, 00:02:15, Serial0/0/1
C       10.0.0.20/30 is directly connected, Serial0/0/0
L       10.0.0.22/32 is directly connected, Serial0/0/0
C       10.0.0.24/30 is directly connected, Serial0/0/1
L       10.0.0.26/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/2428416] via 10.0.0.25, 00:02:18, Serial0/0/1
                        [90/7266816] via 10.0.0.21, 00:02:15, Serial0/0/0
D       172.16.0.192/26 [90/2428416] via 10.0.0.25, 00:02:18, Serial0/0/1
                        [90/7266816] via 10.0.0.21, 00:02:15, Serial0/0/0
D       172.16.1.0/28 [90/2428416] via 10.0.0.25, 00:02:18, Serial0/0/1
                      [90/7266816] via 10.0.0.21, 00:02:15, Serial0/0/0
D    192.168.0.0/23 [90/7264256] via 10.0.0.21, 00:02:04, Serial0/0/0
```

- **D 192.168.0.0/23** y **D 10.0.0.0/28** (resúmenes del lado OSPF) llegan por R-CENTRAL (10.0.0.21) con métrica 7,264,256.
- **D 172.16.0.128/26**, **D 172.16.0.192/26** y **D 172.16.1.0/28** (VLANs de R-EIGRP-1) con dos caminos de métrica distinta: 2,428,416 por el enlace directo (10.0.0.25) y 7,266,816 por R-CENTRAL (10.0.0.21). Es el balanceo de costo desigual (variance).
- **D 10.0.0.16/30** también usa dos caminos (7,264,256 por 10.0.0.21 y 7,520,256 por 10.0.0.25), como efecto de variance sobre toda ruta factible.

3. Evidencia del balanceo de carga en OSPF (costo igual)

Entre R-OSPF-1 y R-OSPF-2 hay dos caminos: el enlace directo L3 y el camino por R-CENTRAL (L1 + L2). Se configuró `ip ospf cost 128` en Serial0/0/1 de ambos routers para que los dos caminos costaran lo mismo.

Costo del enlace directo (L3)

```
R-OSPF-1#show ip ospf interface Serial0/0/1

Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.9/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
```

Costo del enlace hacia R-CENTRAL (L1)

```
R-OSPF-1#show ip ospf interface Serial0/0/0

Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.2/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
```

Las dos rutas de costo igual hacia la misma red

```
R-OSPF-1#show ip route 192.168.0.0
Routing entry for 192.168.0.0/24
Known via "ospf 1", distance 110, metric 129, type intra area
  Last update from 10.0.0.10 on Serial0/0/1, 00:02:12 ago
  Routing Descriptor Blocks:
  * 10.0.0.10, from 2.2.2.2, 00:02:12 ago, via Serial0/0/1
      Route metric is 129, traffic share count is 1
    10.0.0.1, from 2.2.2.2, 00:02:12 ago, via Serial0/0/0
      Route metric is 129, traffic share count is 1
```

Camino de R-OSPF-1 a 192.168.0.0/24	Interfaces de salida	Costo
Directo (por 10.0.0.10)	Se0/0/1 de R-OSPF-1 (128) + Gi0/0 de R-OSPF-2 (1)	**129**
Por R-CENTRAL (por 10.0.0.1)	Se0/0/0 de R-OSPF-1 (64) + Se0/0/1 de R-CENTRAL (64) + Gi0/0 de R-OSPF-2 (1)	**129**

Ambos caminos tienen métrica 129 y `traffic share count is 1`: el tráfico se reparte en partes iguales. En sentido contrario, R-OSPF-2 también tiene dos caminos de costo 129 hacia las VLANs de R-OSPF-1 (sección 2.2).

Evidencia adicional: reparto observado

PC-O1-V10 → PC-O2

```
C:\>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:

Reply from 192.168.0.10: bytes=32 time=1ms TTL=126
Reply from 192.168.0.10: bytes=32 time=2ms TTL=125
Reply from 192.168.0.10: bytes=32 time=1ms TTL=126
Reply from 192.168.0.10: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

Las respuestas alternan entre **TTL=126** (2 routers: camino directo) y **TTL=125** (3 routers: camino por R-CENTRAL). Esto muestra que los dos caminos se usan; Packet Tracer reparte paquete por paquete.

4. Evidencia del balanceo de carga en EIGRP (variance, 75/25)

Se ajustó el retardo de las interfaces (`delay 3000` en el enlace directo L6 y `delay 19900` en los enlaces hacia R-CENTRAL de R-EIGRP-1 y R-EIGRP-2) y se activó `variance 3`. El cálculo completo está en el PDF 1, sección 5.7.

Tabla de topología de R-EIGRP-1

```
R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(4.4.4.4)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/28, 1 successors, FD is 7264256
         via 10.0.0.17 (7264256/2169856), Serial0/0/0
P 10.0.0.16/30, 1 successors, FD is 6752256
         via Connected, Serial0/0/0
P 10.0.0.20/30, 2 successors, FD is 7264256
         via 10.0.0.17 (7264256/2169856), Serial0/0/0
         via 10.0.0.26 (7520256/6752256), Serial0/0/1
P 10.0.0.24/30, 1 successors, FD is 2425856
         via Connected, Serial0/0/1
P 172.16.0.0/25, 2 successors, FD is 2426112
         via 10.0.0.26 (2426112/2816), Serial0/0/1
         via 10.0.0.17 (7264512/2170112), Serial0/0/0
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.10
P 172.16.0.192/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.20
P 172.16.1.0/28, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.99
P 192.168.0.0/23, 1 successors, FD is 7264256
         via 10.0.0.17 (7264256/2169856), Serial0/0/0
```

Rol	Vecino	Métrica total	RD	Verificación
Successor	10.0.0.26 (Se0/0/1, enlace directo)	2,426,112 (= FD)	2,816	Mejor camino
Feasible successor	10.0.0.17 (Se0/0/0, por R-CENTRAL)	7,264,512	2,170,112	RD 2,170,112 < FD 2,426,112 ✓ y 7,264,512 < 3 × FD = 7,278,336 ✓

Proporción: 7,264,512 ÷ 2,426,112 = 2.994 → reparto de **74.96 % por el camino principal y 25.04 % por el secundario**. Packet Tracer muestra "2 successors" porque cuenta ambos caminos instalados en la tabla de enrutamiento; conceptualmente, 10.0.0.26 es el successor y 10.0.0.17 el feasible successor habilitado por `variance`.

Retardo configurado en el enlace hacia R-CENTRAL

```
R-EIGRP-1#show interfaces Serial0/0/0
Serial0/0/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Description: L4 hacia R-CENTRAL Se0/1/0 - 10.0.0.16/30
  Internet address is 10.0.0.18/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 199000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  (... salida recortada ...)
```

`DLY 199000 usec` corresponde a `delay 19900` (el comando se expresa en decenas de microsegundos).

Evidencia adicional: uso del camino secundario

PC-E1-V10 → PC-E2 (20 paquetes)

```
C:\>ping -n 20 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=1ms TTL=126
Reply from 172.16.0.10: bytes=32 time=3ms TTL=125
Reply from 172.16.0.10: bytes=32 time=1ms TTL=126
Reply from 172.16.0.10: bytes=32 time=3ms TTL=125
Reply from 172.16.0.10: bytes=32 time=1ms TTL=126
Reply from 172.16.0.10: bytes=32 time=4ms TTL=125
Reply from 172.16.0.10: bytes=32 time=5ms TTL=126
Reply from 172.16.0.10: bytes=32 time=19ms TTL=125
Reply from 172.16.0.10: bytes=32 time=1ms TTL=126
Reply from 172.16.0.10: bytes=32 time=11ms TTL=125
Reply from 172.16.0.10: bytes=32 time=1ms TTL=126
Reply from 172.16.0.10: bytes=32 time=21ms TTL=125
Reply from 172.16.0.10: bytes=32 time=6ms TTL=126
Reply from 172.16.0.10: bytes=32 time=12ms TTL=125
Reply from 172.16.0.10: bytes=32 time=2ms TTL=126
Reply from 172.16.0.10: bytes=32 time=3ms TTL=125
Reply from 172.16.0.10: bytes=32 time=2ms TTL=126
Reply from 172.16.0.10: bytes=32 time=2ms TTL=125
Reply from 172.16.0.10: bytes=32 time=1ms TTL=126
Reply from 172.16.0.10: bytes=32 time=13ms TTL=125

Ping statistics for 172.16.0.10:
    Packets: Sent = 20, Received = 20, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 21ms, Average = 5ms
```

Las respuestas llegan alternando TTL=126 (camino directo) y TTL=125 (por R-CENTRAL): el camino de costo desigual sí se usa. Packet Tracer alterna los paquetes 1:1 entre los caminos instalados, por eso la proporción 75/25 se demuestra con la configuración y las métricas, como indica el enunciado ("la evidencia es de configuración").

5. Prueba de conectividad de extremo a extremo

Ping y tracert desde **PC-O1-V10** (VLAN 10 de R-OSPF-1, 192.168.1.10) hacia **PC-E1-V10** (VLAN 10 de R-EIGRP-1, 172.16.0.140): de una VLAN de un extremo a una VLAN del extremo opuesto, cruzando de OSPF a EIGRP.

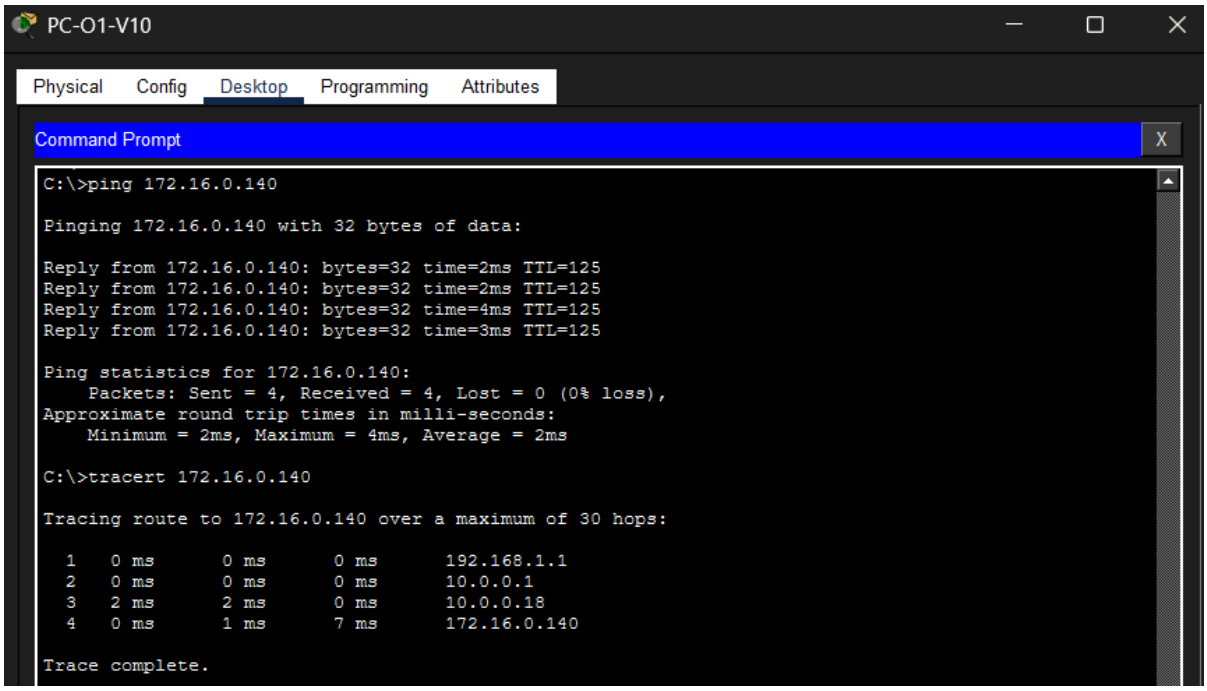


Figura 2. Ping y tracert exitosos de extremo a extremo (PC-O1-V10 → PC-E1-V10).

Salto	Dirección	Equipo
1	192.168.1.1	R-OSPF-1 (Gi0/0.10, gateway de la VLAN 10)
2	10.0.0.1	R-CENTRAL (Se0/0/0): frontera entre OSPF y EIGRP
3	10.0.0.18	R-EIGRP-1 (Se0/0/0)
4	172.16.0.140	PC-E1-V10 (destino)

Resultado: 4 de 4 respuestas, 0 % de pérdida. El TTL=125 (128 − 3 routers) coincide con los 3 routers del tracert.

5.1 Pruebas adicionales (estado final)

PC-O1-V10 (VLAN 10, lado OSPF) → PC-E2 (LAN de R-EIGRP-2)

```
C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=2ms TTL=125
Reply from 172.16.0.10: bytes=32 time=2ms TTL=125
Reply from 172.16.0.10: bytes=32 time=29ms TTL=125
Reply from 172.16.0.10: bytes=32 time=2ms TTL=125

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 29ms, Average = 8ms
```

PC-E1-V20 (VLAN 20 de voz, lado EIGRP) → SRV-O1-V30 (VLAN 30, lado OSPF)

```
C:\>ping 192.168.1.70

Pinging 192.168.1.70 with 32 bytes of data:

Reply from 192.168.1.70: bytes=32 time=2ms TTL=125
Reply from 192.168.1.70: bytes=32 time=2ms TTL=125
Reply from 192.168.1.70: bytes=32 time=2ms TTL=125
Reply from 192.168.1.70: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

PC-E2 (LAN de R-EIGRP-2) → PC-O2 (LAN de R-OSPF-2)

```
C:\>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:

Reply from 192.168.0.10: bytes=32 time=14ms TTL=125
Reply from 192.168.0.10: bytes=32 time=2ms TTL=125
Reply from 192.168.0.10: bytes=32 time=2ms TTL=125
Reply from 192.168.0.10: bytes=32 time=5ms TTL=125

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 14ms, Average = 5ms
```

Anexo A. VLANs y enrutamiento entre VLANs

A.1 Sitio R-OSPF-1

```
SW-OSPF-1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/2
10   DATOS                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10
30   ADMIN_SERVIDORES                 active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
SW-OSPF-1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,30
```

```
R-OSPF-1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.1.1     YES manual up                    up
GigabitEthernet0/0.30  192.168.1.65    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.2        YES manual up                    up
Serial0/0/1            10.0.0.9        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

PC-O1-V10 (VLAN 10) → SRV-O1-V30 (VLAN 30)

```
C:\>ping 192.168.1.70

Pinging 192.168.1.70 with 32 bytes of data:

Reply from 192.168.1.70: bytes=32 time<1ms TTL=127
Reply from 192.168.1.70: bytes=32 time<1ms TTL=127
Reply from 192.168.1.70: bytes=32 time<1ms TTL=127
Reply from 192.168.1.70: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.1.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 192.168.1.70

Tracing route to 192.168.1.70 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.1
  2   1 ms      0 ms      0 ms      192.168.1.70

Trace complete.
```

El tracert muestra al router (192.168.1.1) como primer salto: el tráfico entre VLANs sube por el trunk al router y vuelve a bajar (router-on-a-stick).

A.2 Sitio R-EIGRP-1

```
SW-EIGRP-1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   DATOS                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10
20   VOZ                              active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
99   GESTION                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
SW-EIGRP-1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20,99

SW-EIGRP-1#show ip interface brief
(... salida recortada ...)
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
Vlan99                 172.16.1.2      YES manual up                    up
```

```
R-EIGRP-1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  172.16.0.129    YES manual up                    up
GigabitEthernet0/0.20  172.16.0.193    YES manual up                    up
GigabitEthernet0/0.99  172.16.1.1      YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.18       YES manual up                    up
Serial0/0/1            10.0.0.25       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

PC-E1-V10 (VLAN 10) → IP de gestión del switch (VLAN 99) y → PC-E1-V20 (VLAN 20)

```
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time<1ms TTL=254
Reply from 172.16.1.2: bytes=32 time<1ms TTL=254
Reply from 172.16.1.2: bytes=32 time<1ms TTL=254
Reply from 172.16.1.2: bytes=32 time<1ms TTL=254

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 172.16.0.200

Tracing route to 172.16.0.200 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      172.16.0.129
  2    0 ms      0 ms      0 ms      172.16.0.200

Trace complete.
```

Anexo B. Cableado y vecindades OSPF / EIGRP

B.1 Enlaces seriales de R-CENTRAL

```
R-CENTRAL#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.1        YES manual up                    up
Serial0/0/1            10.0.0.5        YES manual up                    up
Serial0/1/0            10.0.0.17       YES manual up                    up
Serial0/1/1            10.0.0.21       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down

R-CENTRAL#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 64000
(... salida recortada ...)
```

```
R-CENTRAL#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
R-OSPF-1     Ser 0/0/0        154           R       C1900       Ser 0/0/0
R-OSPF-2     Ser 0/0/1        123           R       C1900       Ser 0/0/0
R-EIGRP-1    Ser 0/1/0        179           R       C1900       Ser 0/0/0
R-EIGRP-2    Ser 0/1/1        135           R       C1900       Ser 0/0/0
```

CDP confirma que cada enlace serial llega al router y al puerto planificados.

B.2 OSPF

```
R-CENTRAL#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:37    10.0.0.2        Serial0/0/0
2.2.2.2           0   FULL/  -        00:00:35    10.0.0.6        Serial0/0/1
```

```
R-OSPF-1#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    10.0.0.8 0.0.0.3 area 0
    192.168.1.0 0.0.0.63 area 0
    192.168.1.64 0.0.0.15 area 0
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.30
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:03:21
    2.2.2.2              110      00:03:15
    3.3.3.3              110      00:03:15
  Distance: (default is 110)
```

B.3 EIGRP

```
R-CENTRAL#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.0.0.18       Se0/1/0        10   00:01:32  40     1000  0   5
1   10.0.0.22       Se0/1/1        11   00:01:19  40     1000  0   9
```

```
R-EIGRP-1#show ip protocols
Routing Protocol is "eigrp  100 "
(... salida recortada ...)
  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     10.0.0.16/30
     10.0.0.24/30
     172.16.0.128/26
     172.16.0.192/26
     172.16.1.0/28
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.20
    GigabitEthernet0/0.99
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.17       90            11086320
    10.0.0.26       90            11099826
  Distance: internal 90 external 170
```

Anexo C. Redistribución y sumarización: antes y después

C.1 Antes de la redistribución

R-OSPF-1 solo conoce las redes del dominio OSPF: no hay ninguna ruta 172.16.x. R-CENTRAL, en cambio, conoce los dos dominios (rutas O y D), pero todavía no los comparte entre sí.

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.1, 00:20:48, Serial0/0/0
                    [110/128] via 10.0.0.10, 00:20:48, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
O    192.168.0.0/24 [110/65] via 10.0.0.10, 00:20:48, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30

R-OSPF-1#
```

Figura 3. R-OSPF-1 antes de la redistribución (solo rutas OSPF).

```
R-CENTRAL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/128] via 10.0.0.2, 00:19:31, Serial0/0/0
                    [110/128] via 10.0.0.6, 00:19:31, Serial0/0/1
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.17/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.21/32 is directly connected, Serial0/1/1
D       10.0.0.24/30 [90/2681856] via 10.0.0.18, 00:09:25, Serial0/1/0
                     [90/2681856] via 10.0.0.22, 00:09:12, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
D       172.16.0.0/25 [90/2170112] via 10.0.0.22, 00:09:12, Serial0/1/1
D       172.16.0.128/26 [90/2172416] via 10.0.0.18, 00:09:25, Serial0/1/0
D       172.16.0.192/26 [90/2172416] via 10.0.0.18, 00:09:25, Serial0/1/0
D       172.16.1.0/28 [90/2172416] via 10.0.0.18, 00:09:25, Serial0/1/0
O    192.168.0.0/24 [110/65] via 10.0.0.6, 00:19:31, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.2, 00:19:56, Serial0/0/0
O       192.168.1.64/28 [110/65] via 10.0.0.2, 00:19:56, Serial0/0/0

R-CENTRAL#
```

Figura 4. R-CENTRAL antes de la redistribución: conoce rutas O y D.

C.2 Después de redistribuir, antes de sumarizar

Con `redistribute eigrp 100 metric-type 1 subnets` y `redistribute ospf 1 metric 1544 2000 255 1 1500` en R-CENTRAL, cada dominio recibe las rutas específicas del otro: **O E1** en OSPF y **D EX** (AD 170) en EIGRP. Esta es la evidencia de que la redistribución directa EIGRP → OSPF funcionó antes de reemplazarla por el resumen.

```
R-OSPF-1#show ip route
(... se omite la leyenda de códigos ...)
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.1, 00:00:49, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
O E1    10.0.0.16/30 [110/84] via 10.0.0.1, 00:13:44, Serial0/0/0
O E1    10.0.0.20/30 [110/84] via 10.0.0.1, 00:13:44, Serial0/0/0
O E1    10.0.0.24/30 [110/84] via 10.0.0.1, 00:13:44, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
O E1    172.16.0.0/25 [110/84] via 10.0.0.1, 00:13:44, Serial0/0/0
O E1    172.16.0.128/26 [110/84] via 10.0.0.1, 00:13:44, Serial0/0/0
O E1    172.16.0.192/26 [110/84] via 10.0.0.1, 00:13:44, Serial0/0/0
O E1    172.16.1.0/28 [110/84] via 10.0.0.1, 00:13:44, Serial0/0/0
O    192.168.0.0/24 [110/129] via 10.0.0.10, 00:00:49, Serial0/0/1
                    [110/129] via 10.0.0.1, 00:00:49, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

```
R-EIGRP-1#show ip route
(... se omite la leyenda de códigos ...)
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/2681856] via 10.0.0.17, 00:00:56, Serial0/0/0
D EX    10.0.0.4/30 [170/2681856] via 10.0.0.17, 00:00:56, Serial0/0/0
D EX    10.0.0.8/30 [170/2681856] via 10.0.0.17, 00:00:56, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/0
L       10.0.0.18/32 is directly connected, Serial0/0/0
D       10.0.0.20/30 [90/2681856] via 10.0.0.17, 00:13:25, Serial0/0/0
                     [90/2681856] via 10.0.0.26, 00:13:12, Serial0/0/1
C       10.0.0.24/30 is directly connected, Serial0/0/1
L       10.0.0.25/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 7 subnets, 4 masks
D       172.16.0.0/25 [90/2170112] via 10.0.0.26, 00:13:12, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/26 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
C       172.16.1.0/28 is directly connected, GigabitEthernet0/0.99
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0.99
D EX 192.168.0.0/24 [170/2681856] via 10.0.0.17, 00:00:56, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
D EX    192.168.1.0/26 [170/2681856] via 10.0.0.17, 00:00:56, Serial0/0/0
D EX    192.168.1.64/28 [170/2681856] via 10.0.0.17, 00:00:56, Serial0/0/0
```

C.3 Rebote del resumen por la redistribución mutua

Al crear el resumen EIGRP 192.168.0.0/23 mientras se redistribuía EIGRP hacia OSPF, ese resumen volvió al dominio OSPF como `O E1 192.168.0.0/23`. Era inofensivo (las rutas específicas de las VLANs ganan) y desapareció al reemplazar la redistribución de EIGRP por la del resumen estático.

```
R-OSPF-1#show ip route
(... salida recortada: solo rutas externas ...)
O E1    10.0.0.16/30 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
O E1    10.0.0.20/30 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
O E1    10.0.0.24/30 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
O E1    172.16.0.0/25 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
O E1    172.16.0.128/26 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
O E1    172.16.0.192/26 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
O E1    172.16.1.0/28 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
O E1 192.168.0.0/23 [110/84] via 10.0.0.1, 00:02:33, Serial0/0/0
```

C.4 Comandos de redistribución y sumarización en R-CENTRAL

```
! R-CENTRAL - redistribución y sumarización (configuración final)
router eigrp 100
 eigrp router-id 3.3.3.3
 redistribute ospf 1 metric 1544 2000 255 1 1500
 network 10.0.0.16 0.0.0.15
!
router ospf 1
 router-id 3.3.3.3
 redistribute static metric-type 1 subnets
 network 10.0.0.0 0.0.0.15 area 0
!
interface Serial0/1/0
 ip summary-address eigrp 100 192.168.0.0 255.255.254.0
 ip summary-address eigrp 100 10.0.0.0 255.255.255.240
interface Serial0/1/1
 ip summary-address eigrp 100 192.168.0.0 255.255.254.0
 ip summary-address eigrp 100 10.0.0.0 255.255.255.240
!
ip route 172.16.0.0 255.255.254.0 Null0
ip route 10.0.0.16 255.255.255.240 Null0
```

Anexo D. Limitaciones de Packet Tracer

Salidas que justifican las soluciones alternativas descritas en el PDF 1, sección 6.

Orden de palabras rechazado (solución: metric-type 1 subnets)

```
R-CENTRAL(config-router)#redistribute eigrp 100 subnets metric-type 1
                                                              ^
% Invalid input detected at '^' marker.
```

summary-address no existe en OSPF; el resumen de EIGRP sí existe

```
R-CENTRAL(config)#router ospf 1
R-CENTRAL(config-router)# summary-address ?
% Unrecognized command
R-CENTRAL(config-router)# exit
R-CENTRAL(config)#interface Serial0/1/0
R-CENTRAL(config-if)# ip summary-address ?
  eigrp  Enhanced Interior Gateway Routing Protocol (EIGRP)
```

Sin herramientas de filtrado (distribute-list, route-map, prefix-list)

```
R-CENTRAL(config-router)# redistribute eigrp 100 ?
  metric       Metric for redistributed routes
  metric-type  OSPF/IS-IS exterior metric type for redistributed routes
  subnets      Consider subnets for redistribution into OSPF
  tag          Set tag for routes redistributed into OSPF
  <cr>
R-CENTRAL(config-router)# distribute-list ?
% Unrecognized command
R-CENTRAL(config-router)# exit
R-CENTRAL(config)#route-map ?
% Unrecognized command
R-CENTRAL(config)#ip prefix-list ?
% Unrecognized command
```